ETALE SEPARATED MORPHISMS COMMUTE WITH THE WELL-NIGH-AFFINE MODULI SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3691-3708 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3710-3795, the complete author proof (84 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_stacks-stacks-more-morphisms.json (pb=3538, stmt 3691-3708, proof 3710-3795).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 03JE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-moduli-space-finite-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3592–3599. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine-moduli-space: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3425–3433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine-moduli-space-etale: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3517–3533. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07S0: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MQ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JA: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0418: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DU9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DU7: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DTW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3795–3795. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $h : \mathcal{X}' \to \mathcal{X}$ be a morphism of algebraic stacks. Assume $\mathcal{X}$ is well-nigh affine, h is étale, h is separated, and h induces isomorphisms on automorphism groups (Morphisms of Stacks, Remark 0DTW). Then there exists a cartesian diagram*

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ M' & \longrightarrow & M \end{array}$$

where $M' \to M$ is a separated étale morphism of schemes and $\mathcal{X} \to M$ is the moduli space constructed in Lemma lemma-well-nigh-affine-moduli-space.

Proof. Choose an affine scheme U and a surjective, flat, finite, and locally finitely presented morphism $U \to \mathcal{X}$. Since h is representable by algebraic spaces (Morphisms of Stacks, Lemmas 0DU9 and 0DU7) we see that $U' = \mathcal{X}' \times_{\mathcal{X}} U$ is an algebraic space. Since $U' \to U$ is separated and étale, we see that U' is a scheme and that every finite set of points of U' is contained in an affine open, see Morphisms of Spaces, Lemma 0418 and More on Morphisms, Lemma 07S0. Setting

$R' = U' \times_{\mathcal{X}'} U'$ we see that $s', t' : R' \to U'$ are finite locally free. By Groupoids, Lemma 03JE there exists an open covering $U' = \bigcup U'_i$ by R'-invariant affine open subschemes $U'_i \subset U'$. Let $\mathcal{X}'_i \subset \mathcal{X}'$ be the corresponding open substacks. These are well-nigh affine as $U'_i \to \mathcal{X}'_i$ is surjective, flat, finite and of finite presentation. By Lemma lemma-well-nigh-affine-moduli-space-etale we obtain cartesian diagrams

$$\begin{array}{ccc} \mathcal{X}'_i & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ M'_i & \longrightarrow & M \end{array}$$

with $M'_i \to M$ an étale morphism of affine schemes and vertical arrows as in Lemma lemma-well-nigh-affine-moduli-space. Observe that $\mathcal{X}'_{ij} = \mathcal{X}'_i \cap \mathcal{X}'_j$ is an open subspace of $\mathcal{X}'_i$ and $\mathcal{X}'_j$. Hence we get corresponding open subschemes $V_{ij} \subset M'_i$ and $V_{ji} \subset M'_j$. By the result of Lemma lemma-moduli-space-finite-affine we see that both $\mathcal{X}'_{ij} \to V_{ij}$ and $\mathcal{X}'_{ji} \to V_{ji}$ are categorical moduli spaces! Thus we get a unique isomorphism $\varphi_{ij} : V_{ij} \to V_{ji}$ such that

$$\begin{array}{ccccccc} \mathcal{X}'_i & \longleftarrow & \mathcal{X}'_i \cap \mathcal{X}'_j & & & \longrightarrow & \mathcal{X}'_j \\ \downarrow & & \downarrow & & & & \downarrow \\ M'_i & \longleftarrow & V_{ij} & \xrightarrow{\varphi_{ij}} & V_{ji} & \longrightarrow & M'_j \end{array}$$

is commutative. These isomorphisms satisfy the cocyclce condition of Schemes, Section 01JA by a computation (and another application of the previous lemma) which we omit. Thus we can glue the affine schemes in to scheme M', see Schemes, Lemma 01JB. Let us identify the M'_i with their image in M'. We claim there is a morphism $\mathcal{X}' \to M'$ fitting into cartesian diagrams

$$\begin{array}{ccc} \mathcal{X}'_i & \longrightarrow & \mathcal{X}' \\ \downarrow & & \downarrow \\ M'_i & \longrightarrow & M' \end{array}$$

This is clear from the description of the morphisms into the glued scheme M' in Schemes, Lemma 01JB and the fact that to give a morphism $\mathcal{X}' \to M'$ is the same thing as given a morphism $T \to M'$ for any morphism $T \to \mathcal{X}'$. Similarly, there is a morphism $M' \to M$ restricting to the given morphisms $M'_i \to M$ on M'_i. The morphism $M' \to M$ is étale (being étale on the members of an étale covering) and the fibre product property holds as it can be checked on members of the (affine) open covering $M' = \bigcup M'_i$. Finally, $M' \to M$ is separated because the composition $U' \to \mathcal{X}' \to M'$ is surjective and universally closed and we can apply Morphisms, Lemma 09MQ. $\square$